

**1 (a)**

$$R = \frac{V}{I} \text{ and } R = \frac{\rho L}{A}, \text{ hence } \frac{V}{I} = \frac{\rho L}{A} \Rightarrow \rho = \frac{VA}{IL} = \frac{\pi Vd^2}{4IL}$$

**(b)**

$$\rho = \frac{\pi Vd^2}{4IL} = \frac{1.50 \times \pi \times (0.48 \times 10^{-3})^2}{4 \times 0.82 \times 0.632} = 5.23 \times 10^{-7} \Omega \text{ m}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta V}{V} + \frac{\Delta I}{I} + \frac{\Delta L}{L} + 2 \frac{\Delta d}{d} = \frac{0.01}{1.50} + \frac{0.01}{0.82} + \frac{0.1}{63.2} + 2 \frac{0.02}{0.48} = 0.1037$$

$$\Delta \rho = 0.1037 \rho = 0.1037 \times 5.23 \times 10^{-7} = 0.54 \times 10^{-7} \approx 0.5 \times 10^{-7} \Omega \text{ m}$$

$$\text{Hence, } \rho = (5.2 \pm 0.5) \times 10^{-7} \Omega \text{ m}$$

**2 (a)**

Conditions for equilibrium: